

EEE554 – Probability and Random Processes

Module 4: Independence and Repeated Experiments

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From Conditional Probability to Independence

In the previous module, we saw that learning that B occurred generally changes the probability of A :

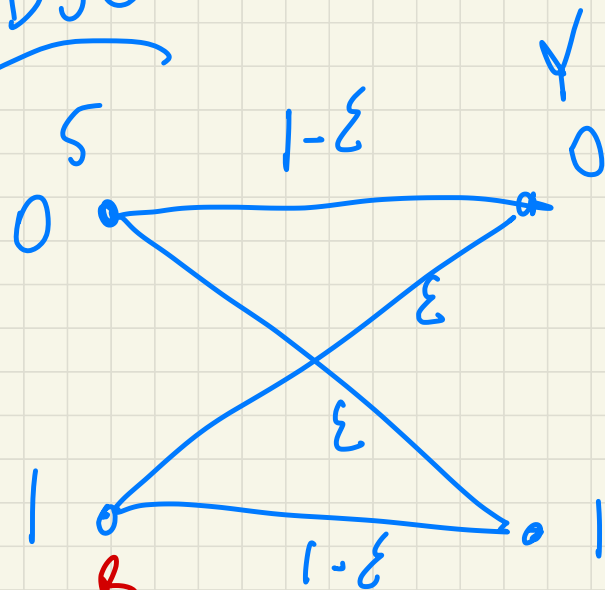
$$P(A|B) \neq P(A).$$

But what if learning that B occurred provides no information about A ? That is,

$$\boxed{P(A|B) = P(A).}$$

In this case, A and B are said to be *statistically independent*.

BSC



$$\varepsilon = 1/2$$

$$P(S=0) = P(S=1) = \frac{1}{2}$$

$$P(\overset{A}{Y=1} | \overset{B}{S=0}) = 0.5, \quad P(Y=1 | S=1) = 0.5$$

$$P(\overset{A}{Y=1}) = P(Y=1 | S=1) \cdot 0.5 + P(Y=1 | S=0) \cdot 0.5 = 0.5$$

Why Independence?

Many engineering systems involve repeated experiments.

Examples:

- Multiple coin tosses
- Packet transmissions
- Successive channel uses
- Sensor measurements

Question:

Does the occurrence of one event affect the probability of another?

Independent Events

Definition 1. Events A and B are statistically independent if

$$P(A \cap B) = P(A)P(B).$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

If $P(B) > 0$, this is equivalent to

$$P(A|B) = P(A). \quad ; \quad P(B|A) = P(B)$$

Knowing whether B occurred does not change the probability of A .

If $P(A) > 0$ and $P(B) > 0$,

$$P(A|B) = P(A) \iff P(B|A) = P(B).$$

Thus, independence is a symmetric relationship.

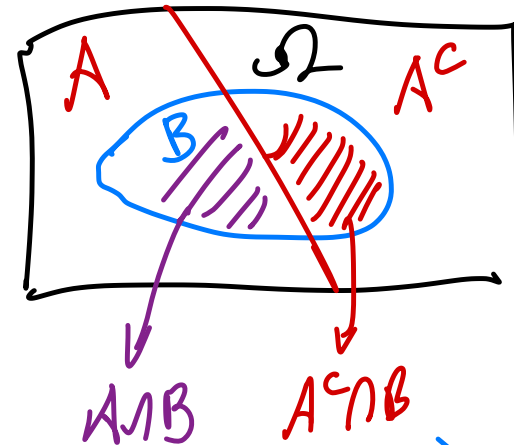
Complements

If $A \perp B$, then

$$\rightarrow A^c \perp B,$$

$$A \perp B^c,$$

$$A^c \perp B^c.$$



$$P(A^c \cap B) + P(A \cap B) = P(B)$$

$\nmid A \perp B: P(A) \cdot P(B)$

$$P(A^c \cap B) = P(B) - P(A) \cdot P(B) = P(B) \cdot (1 - P(A))$$

$= P(B) \cdot P(A^c)$

Mutual Independence

Events $A_1, \dots, A_n$ are mutually independent if

$$P\left(\bigcap_{i \in I} A_i\right) = \prod_{i \in I} P(A_i)$$

for every subset

$$I \subseteq \{1, \dots, n\}, \quad |I| \geq 2.$$

Ex: A_1, A_2 :

$$P(A_1 \cap A_2) = P(A_1) \cdot P(A_2)$$

Ex: A_1, A_2, A_3

$$P(A_1 \cap A_2 \cap A_3) = P(A_1) \cdot P(A_2) \cdot P(A_3)$$

$$P(A_1 \cap A_2) = P(A_1) \cdot P(A_2)$$

$$P(A_1 \cap A_3) = P(A_1) \cdot P(A_3); \quad P(A_2 \cap A_3) = P(A_2) \cdot P(A_3)$$

Ex A_1, A_2, A_3, A_4

$$P(A_1 \cap A_2 \cap A_3 \cap A_4) = \dots$$

$$P(A_1 \cap A_2), P(A_1 \cap A_3), P(A_1 \cap A_4) \dots$$

$$\parallel$$
$$P(A_1) \cdot P(A_2)$$

$$P(A_1 \cap A_2 \cap A_3), P(A_1 \cap A_2 \cap A_4), P(A_2 \cap A_3 \cap A_4)$$

Pairwise vs. Mutual Independence

Events $A_1, \dots, A_n$ are *pairwise independent* if

$$P(A_i \cap A_j) = P(A_i)P(A_j), \quad i \neq j.$$

Pairwise independence does not imply mutual independence.

Takeaway:

Checking independence two events at a time is *not* sufficient to guarantee mutual independence.

Example: Pairwise Independence $\nRightarrow$ Mutual Independence

Experiment: Flip two fair coins.

$$\Omega = \{HH, HT, TH, TT\},$$

with $P(\omega) = \frac{1}{4}$, $\forall \omega \in \Omega$. Define the events

$$A = \{\text{first toss is H}\} = \{HH, HT\},$$

$$B = \{\text{second toss is H}\} = \{HH, TH\},$$

$$C = \{\text{both tosses are equal}\} = \{HH, TT\}.$$

Observe that

$$P(A) = P(B) = P(C) = \frac{1}{2}.$$

Moreover,

$$P(A \cap B) = P(A \cap C) = P(B \cap C) = \frac{1}{4} = \frac{1}{2} \cdot \frac{1}{2}.$$

Hence, A , B , C are pairwise independent.

However,

$$A \cap B \cap C = \{HH\},$$

so that

$$P(A \cap B \cap C) = \frac{1}{4} \neq \frac{1}{8} = P(A)P(B)P(C).$$

Therefore,

A, B, C are pairwise independent, but not mutually independent.

Independent Repeated Experiments

A particularly important application of independence occurs when an experiment is repeated. Suppose an experiment is repeated n times.

For the i -th experiment, let

$$(\Omega_i, \mathcal{F}_i, P_i)$$

be its probability space.

An outcome is

$$\omega = (\omega_1, \omega_2, \dots, \omega_n), \quad \omega_i \in \Omega_i.$$

The sample space of the combined experiment is

$$\Omega = \Omega_1 \times \Omega_2 \times \dots \times \Omega_n.$$

Die roll: nsz

$$\{1, \dots, 6\} \times \{1, \dots, 6\}$$

$$= \{(1,1), (1,2), \dots, (1,6), (2,1), \dots, (2,6), \dots, (6,6)\}$$

Suppose

$$A_i \in \mathcal{F}_i$$

is an event associated with the i -th experiment.

The event that

$$A_1, \dots, A_n$$

all occur is represented in the product space by

$$A_1 \times A_2 \times \dots \times A_n.$$

If the experiments are independent, then

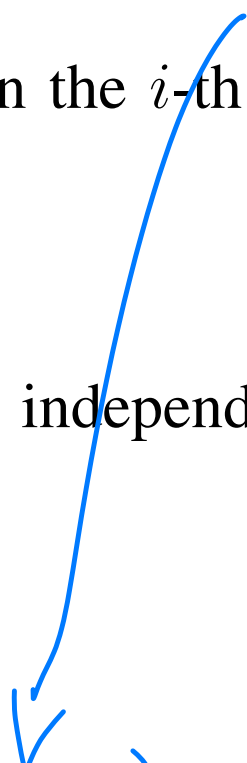
$$P(A_1 \times \cdots \times A_n) = \prod_{i=1}^n P_i(A_i).$$

Equivalently, if E_i denotes the event in the joint sample space that A_i occurs in the i -th experiment, then

$$A_1 \times \cdots \times A_n = \bigcap_{i=1}^n E_i.$$

Hence, independence can also be written as

$$P\left(\bigcap_{i=1}^n E_i\right) = \prod_{i=1}^n P(E_i).$$


$$E_i = \{(\omega_1, \dots, \omega_n) \in \Omega_1 \times \Omega_2 \times \cdots \times \Omega_n : \omega_i \in A_i\}$$

Identical Repeated Experiments

If the same experiment is repeated n times, then

$$\Omega_1 = \cdots = \Omega_n = \Omega,$$

and the joint sample space becomes

$$\boxed{\Omega^n = \underbrace{\Omega \times \cdots \times \Omega}_{n \text{ times}}.}$$

An outcome is

$$(\omega_1, \dots, \omega_n).$$

Bernoulli Trials

Single trial:

$$\Omega = \{S, F\},$$

where $P(S) = p$, $P(F) = 1 - p$.

Repeat independently n times. Sample space:

Ω^n . $\rightarrow 2^n$ joint outcomes

Question: What is the probability of observing exactly k successes,

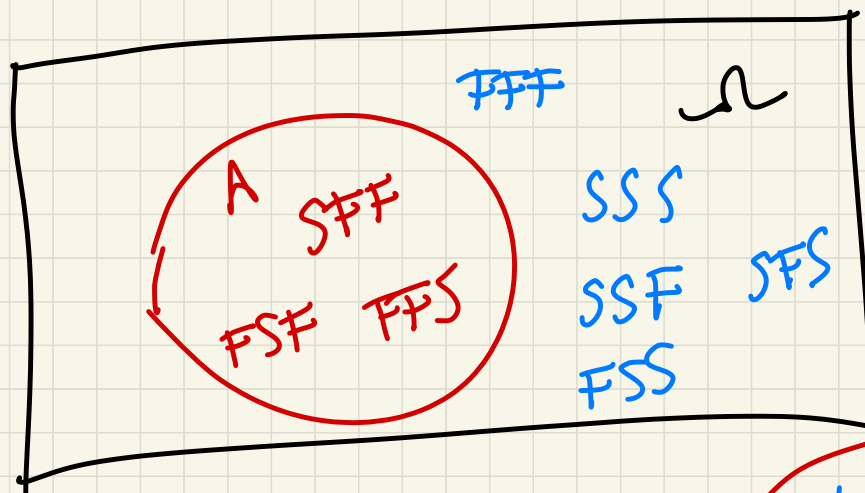
$P_n(k)$?

Ex

$K=1$

$n=3$

$A \equiv \{ \text{only 1 success} \}$



$$= 3 \cdot p \cdot (1-p)^2$$

$$P(A) = P(\{SFF, FSF, FFS\}) = P(SFF) + P(FSF) + P(FFS)$$

$$P(S \neq F)$$

||

$$P(A_1 \times A_2 \times A_3)$$

$$= P(A_1) \cdot P(A_2) \cdot P(A_3)$$

$$\rightarrow P(w_1 = S) \cdot P(w_2 = F) \cdot P(w_3 = F) = p(1-p)^2$$

$$P(FS\neq) = P(F\neq S) = p(1-p)^2$$

$$A_1 = \{w_1 = S\}$$

$$A_2 = \{w_2 = F\}$$

$$A_3 = \{w_3 = F\}$$

Binomial Distribution

Suppose exactly k of the n trials are successes.

Any particular sequence containing k successes and $n - k$ failures has probability

$\underbrace{\times k}_{SSS \sim S} \quad \underbrace{\times n-k}_{FFF \sim F}$

$$p^k(1 - p)^{n-k}.$$

There are

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

such sequences.

Therefore,

$$P_n(k) = \binom{n}{k} p^k (1-p)^{n-k}.$$

A handwritten blue ink derivation of the binomial coefficient formula. A blue arrow points from the $\binom{n}{k}$ term in the boxed equation above to the expression $\frac{n!}{k! (n-k)!}$. This is followed by an equals sign and a fraction: the numerator is $n \cdot (n-1) \cdot \dots \cdot (n-k+1)$ (with an ellipsis and a curved line over the terms), and the denominator is $k \cdot (k-1) \cdot \dots \cdot 1$ (with an ellipsis and a curved line over the terms).

$$\frac{n!}{k! (n-k)!} = \frac{n \cdot (n-1) \cdot \dots \cdot (n-k+1)}{k \cdot (k-1) \cdot \dots \cdot 1}$$

Communication Example

Binary Symmetric Channel

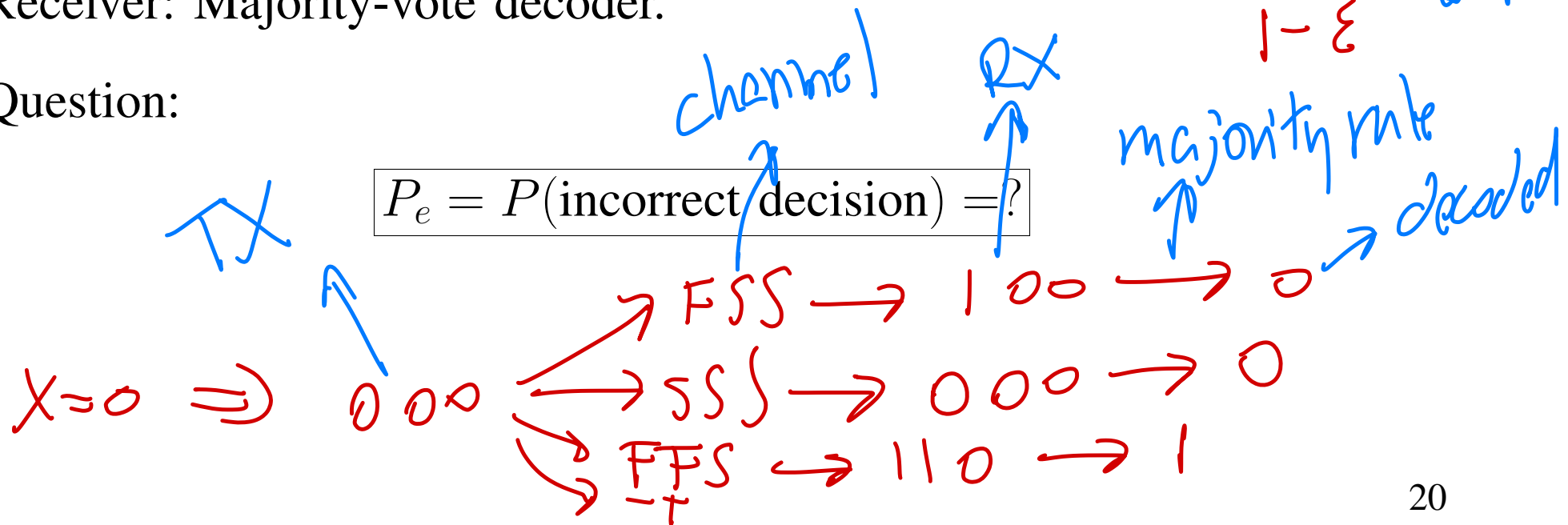
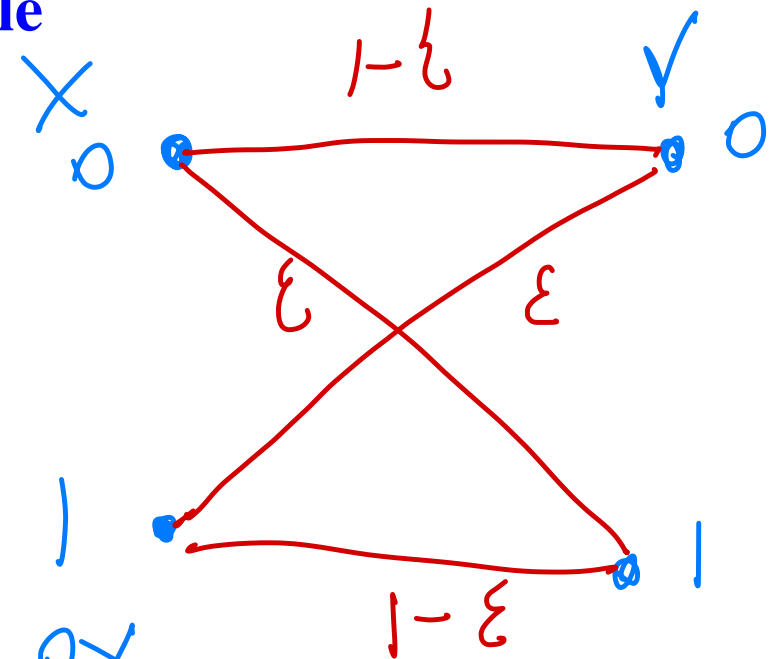
$$P(\text{bit error}) = \varepsilon.$$

Transmit each bit 3 times independently.

Receiver: Majority-vote decoder.

Question:

$$P_e = P(\text{incorrect decision}) = ?$$



$$P_e = P(\text{channel introduces 2 or 3 errors})$$

An incorrect decision occurs if and only if (iff) there are exactly 2 or 3 errors in the sequence:

$$= P(A_2 \cup A_3), \quad A_2 \cap A_3 = \emptyset$$

$$A_2 = \{\text{exactly two errors}\}$$

$$A_3 = \{\text{exactly three errors}\}.$$

Then:

$$P_e = P(A_2) + P(A_3)$$

Exactly two errors:

$$P(A_2) = \binom{3}{2} \underbrace{\varepsilon^2(1-\varepsilon)}_{P(\text{FFS})} = 3 \cdot \varepsilon^2(1-\varepsilon)$$

Exactly three errors:

$$P(A_3) = \binom{3}{3} \epsilon^3. \quad \text{SSS} \quad \epsilon^3$$

Therefore,

$$P_e = 3\epsilon^2(1 - \epsilon) + \epsilon^3.$$

For example, if an individual error occurs with probability $\epsilon = 0.1$:

$$P_e = 3(10^{-1})^2(1 - 10^{-1}) + (10^{-1})^3 \approx 2.8 \times 10^{-2}.$$

By exploiting repetitions, the probability of incorrect bit detection can be significantly lowered.

Important:

The improvement relies on the errors being independent.

If the same disturbance causes all three transmissions to fail simultaneously, repetition may provide little or no benefit.

Takeaway

Repeated independent experiments naturally lead to

- Cartesian products,
- product probability spaces,
- Bernoulli trials,
- the Binomial distribution.

These concepts form the mathematical foundation of many engineering systems, including communication, coding, networking, and machine learning.